Lab session 2: Diode

1. OBJECTIVE OF THIS LAB SESSION

The second lab session in electronics focuses on the electronic component diode.

Please define yourself why diodes areimportant to you based on current research. Please cite min 3 papers from 2025

Part 1: Characteristic curves of germanium and silicon diode

Theory

The transmission characteristics $I_F = f(U_F)$ of a germanium diode AA138 and a silicon diode 1N4148 are suitable for currents $I_F = 0..10mA$ with the X-Y mode of the oscilloscope

Preparation 1

- Draw a circuit diagram to plot the diode current on the Y-axis and the voltage drop on the diode on the X-axis.
- Explain why this setting is also suitable to ensure safe operation of the diode, even if e.g. the diode temperature rises.
- Find the dimensions & important characteristic values for both diodes in the corresponding data-sheets.

Lab-Task 1:

- Plot the forward characteristics of both diodes on one sheet based on simulated and imported oscilloscope data

Evaluation 1

- Graphically determine the differential (AC) resistances at $I_F = 2mA$ from the graphs.

Part 2: Characteristic curve of the silicon diode in semilogarithmic representation

Theory

A logarithmic representation of the characteristic curve allows you to determine several characteristic values, i.e. the *reverse saturation current* I_S, *bulk resistance* R_B and the e*mission coefficient* n of the diode.

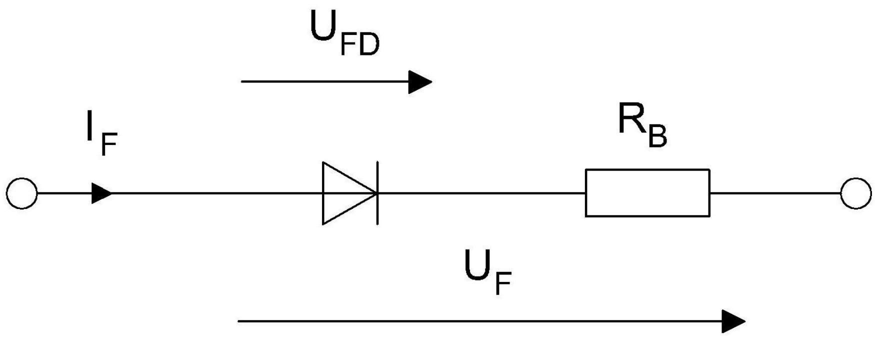

Figure: Equivalent circuit diagram for a diode

The equivalent circuit of the diode helps to derive a relation for forward voltage and forward current of the diode $U_F = f(I_F)$

$$U_F = U_{FD} + I_F \cdot R_B$$

using the Shockley equation for the ideal diode

$$I_F = I_S e^{\frac{U_{FD}}{n \cdot U_T}}$$

and taking into account the impact of the bulk resistance R_B by eliminating the unknown quantity U_{FD}.

Preparation 2

- Summarize the formulas to derive the above-mentioned quantities
- Derive the relation and calculate the theoretic differential resistance r_D.

Lab-Task 2:

- Measure the forward characteristic $I_F = f(U_F)$ of the silicon diode 1N4148 in a current range of $10\mu A \cdots 100mA$. Subdivide decade into 3 to 4 values, e.g. in the sequence $10\mu A, 20\mu A, 50\mu A, 100\mu A, 200\mu A \ldots$.

Evaluation 2

- Compare the differential resistance r_D at $I_F = 2mA$ using the relation $U_F = f(I_F)$ derived as preparation and compare with the actual value determined in Exercise 1.

- Use Excel, Matlab or libre office for the graphic representation.

Example:

```
u_f=[0.4,0.43,0.47,0.5,0.54,0.58,0.62,0.65,0.7,0.74,0.78,0.83,0.88]% (generates number vector for U_F)
i_f=[10e-6,20e-6,50e-6,100e-6,200e-6,500e-6,1e-3,2e-3,5e-3,10e-3,20e-3,50e-3,100e-3]% (generates number vector for I_F)
figure % (opens a new plot window)
plot(u_f,i_f)% (plots u_f on linear scale and i_f on linear scale)
figure% (opens a new plot window)
semilogy(u_f,i_f)% (plots u_f on linear scale and i_f on logarithmic scale)
```

- Plot the measured values on a logarithmic scale and
- graphically determine all above-mentioned quantities.

Also, with MATLAB, the diode characteristic can also be calculated using the relation $U_F = f(I_F)$ derived as preparation from the measured values of the current (numerical vector i_f).

Modify your code accordingly, and

```
u_f=[0.4,0.43,0.47,0.5,0.54,0.58,0.62,0.65,0.7,0.74,0.78,0.83,0.88]% (generates number vector for U_F)
i_f=[10e-6,20e-6,50e-6,100e-6,200e-6,500e-6,1e-3,2e-3,5e-3,10e-3,20e-3,50e-3,100e-3]% (generates number vector for I_F)

m=%your determined value for the emission coefficient
i_s=%your determined value for the reverse saturation current
Rb=%your determined value for the bulk resistance

u_fTH=%%put your Formula here%%

figure % (opens a new plot window)
plot(u_f,i_f,u_fTH,i_f)% (plots u_f vs. i_f and the theoretic curve on linear-linear scales)
figure% (opens a new plot window)
semilogy(u_f,i_f,u_fTH,i_f)% (plots u_f vs. i_f and the theoretic curve on linear-logarithmic scales)
```

- plot the theoretic curve in the same diagram as the measured one.

Part3: Half-wave rectifier circuit

Theory

In this part of the lab session the ripple factor of the output voltage in a half-wave rectifier circuit is investigated by using different capacitors. The ripple factor is an indication of the effectiveness of the rectification and defined as the ratio of the RMS-value of the rectified signal to its mean value.

Preparation 3

- Derive a formula for the ripple factor as a function of the TRMS value as well as the RMS value of the alternating part only.

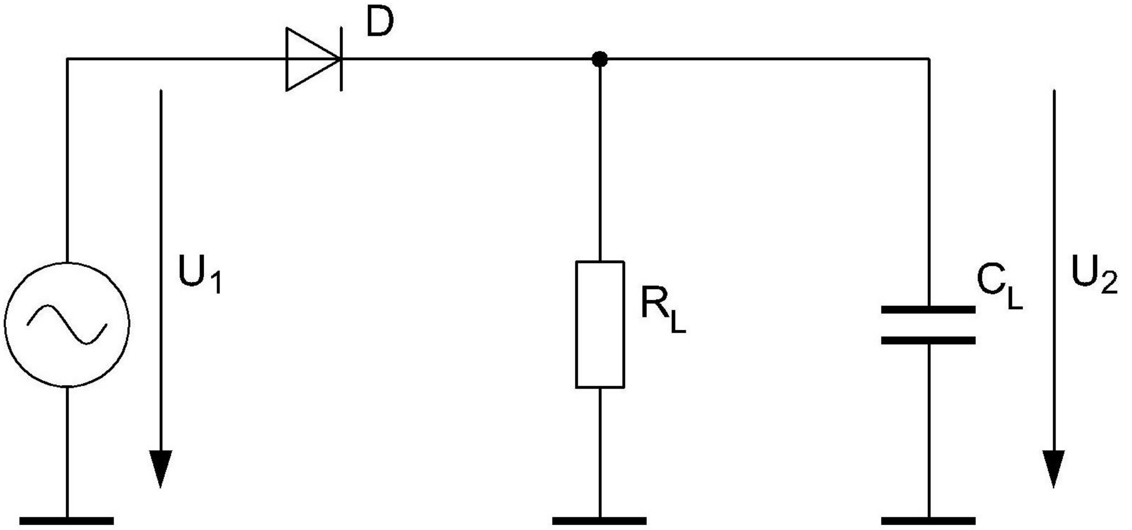

Figure: Half-wave rectifier circuit to investigate the ripple factor.
$R_L = 10k\Omega$, $U_1 = 10V$ (peak value), $C_L = 47nF$ and $C_L = 470nF$, *f=1kHz. As diode use the 1N4148*

Lab-Task 3:

- Set-up the half-wave rectifier circuit illustrated in the figure above
- Use an oscilloscope to display and record the voltage curves of U_1 and U_2 for the two cases $C_L = 47nF$ and $C_L = 470nF$
- Use the digital multimeter to measure the trueRMS value, the RMS value of the alternating part only and the mean value of U_1 and U_2 for the two cases $C_L = 47nF$ and $C_L = 470nF$

Evaluation 3

- Determine the ripple factor of U_2 for both cases.

Part 4: Full-wave rectifier circuit

Theory

In this part of the lab session the ripple factor of the output voltage in a full-wave rectifier circuit is investigated by using different capacitors. The ripple factor is an indication of the effectiveness of the rectification and defined as the ratio of the RMS-value of the rectified signal to its mean value.

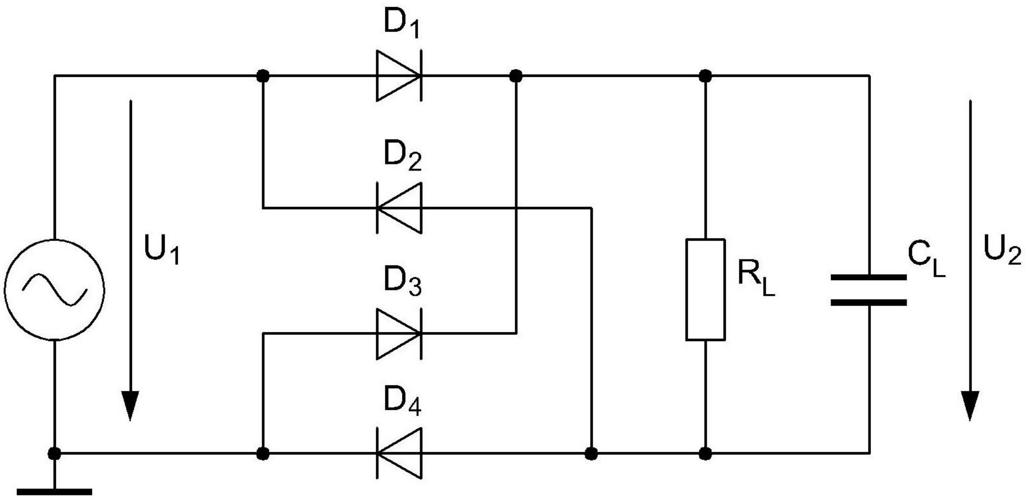

Figure: Full-wave rectifier circuit to investigate the ripple factor.
$R_L = 10k\Omega$, $U_1 = 10V$ *(peak value)*, $C_L = 47nF$ *and* $C_L = 470nF$, *f=1kHz. As diode use the 1N4148*

Preparation 4

- The oscilloscopes input channels (and its probes) are grounded, same as output connector of the signal generator. Please explain why you cannot directly measure the rectified voltage by an oscilloscope but have to use a “differential probe”.

Lab-Task 4:

- Set-up the half-wave rectifier circuit illustrated in the figure above.
- Use an oscilloscope to display and record the voltage curves of U_1 and U_2 for the two cases $C_L = 47nF$ and $C_L = 470nF$
- Use the digital multimeter to measure the trueRMS value, the RMS value of the alternating part only and the mean value of U_1 and U_2 for the two cases $C_L = 47nF$ and $C_L = 470nF$

Evaluation 4

- Determine the ripple factor of U_2 for both cases.